

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONER
RAIGAD -402 103

End Semester Examination – Summer, May – 2019

Course: B. Pharm

Sem.: - IV

Subject with Subject Code: -Pharmacognosy & Phytochemistry I (BP405T)

Date: -06/06/2019

Time: -3 Hrs.

Marks: 75

Instructions:

- I. All questions are compulsory
- II. Illustrate your answers with neat sketches, diagram etc., wherever necessary

Q.No.1. Multiple choice questions

(20×1=20) (Marks)

1. What is the composition of Dragendorff's Reagent?
 - a) Potassium bismuth iodide
 - b) Potassium mercuric iodide
 - c) Solution of iodine in potassium iodide
 - d) Saturated solution of picric acid
2. The fundamental principle of homeopathy is
 - a) Disease can be treated by antibiotics
 - b) Unlike can be treated by like
 - c) like can be treated by like
 - d) none of the above
3. Hemp fibres are obtained from?
 - (i) *Cannabis sativa*
 - (ii) *Ephedra gerardiana*
 - (iii) *Swertiachirata*
 - (iv) *Gossypiumbarbadense*
4. Important chemical constituent of honey is
 - a) Glucose
 - b) Fructose
 - c) Maltose
 - d) Invert sugar
5. Which of the following is not unorganized drug
 - (a) Tragacanth
 - (b) Lard
 - (c) Aloe
 - (d) Datura
6. Under which classification the crude drugs are classified on the basis of their therapeutic uses.
 - a) Chemical
 - b) Alphabetical
 - c) Pharmacological
 - d) Morphological
7. 1mg of lycopodium powder contains _____ spores.
 - a) 94000
 - b) 93000
 - c) 84000
 - d) 74000
8. Pest is a
 - a) Undesired animal
 - b) Undesired plant
 - c) both A & B
 - d) None of the above
9. Micro propagation of the plant is carried out by
 - a) Seed germination
 - b) Cross fertilization
 - c) Grafting
 - d) Tissue culture

10. Polyploidy is defined as

- i) Addition of chromosomes
- ii) Multiplication of entire chromosomes set
- iii) submicroscopic changes in DNA material
- iv) Cross structural changes

a) Only A b) Only B c) A, B & C d) all of the above

11) Coppicing is the method used for collection of _____

- a) Roots b) Stems c) Bark d) Leaves

12) Surface sterilization is not done by _____

- a) Sodium hypochloride (2%) b) Calcium hypochloride (9-10%)
- c) Mercuric chloride (0.1-1%) d) Sodium chloride (0.9%)

13) An explant is

- a) Excised piece of tissue or organ used for culture
- b) A organ meant for respiration
- c) A secondary metabolite
- d) none of the above

14) Goldbeater's skin test is used to identify _____

- a) Flavonoids b) Tannins
- b) Resins d) Cynogenetic glycosides

15) _____ is the identification test for flavonoids

- i) Phenazone test ii) Lead acetate test
- iii) Shinodaest iv) Match stick test
- a) i and iv b) ii and iii c) i and iii d) ii and iv

16) Which of the following is not present in jute _____

- a) Hemicellulose b) True cellulose
- c) Lignin d) Resin

17) Total ash of crude drug signifies

- (i) Organic content of the drug
- (ii) Mineral matter in drug
- (iii) Addition of extraneous matter such as sand & stone etc.
- (iv) Woody matter present in the drug

a) i and ii b) ii and iii c) i and iii d) ii and iv

18) which of the following is biological source of Beeswax

- (a) *Apis mellifera* (b) *Ovis aries*
- (c) *Ananas comosus* (d) None of the above

19) Which of the following is indole alkaloid?

20) Which of the following is not the property of volatile oil.

- (a) They evaporates at room temperature
- (b) They are optically active
- (c) Most of them are heavier than water
- (d) They possesses high refractive indices

Q.No. 2 Attempt any Two of the following (Long Answers) (2×10=20) Marks

- a.) Define & classify Alkaloids. Write the physical & chemical properties of Alkaloids with chemical test.
- b.) Enlist various methods for evaluation of crude drugs. Explain Physical, chemical & microscopical methods of evaluation.
- c.) Enlist different factors affecting cultivation. Explain different method for cultivation & collection of crude drugs.

Q.No .3 Attempt any seven of the following (7 ×5=35) Marks

- a.) Explain applications of plant tissue culture.
- b.) Write a note on Ayurveda.
- c.) Explain in details about Plant growth regulators.
- d.) Define & Classify Glycosides with example. Write its physical & chemical properties.
- e.) Enlist types of culture. Explain Protoplast culture.
- f.) Define Pharmacognosy. Enlist different sources of drugs with suitable example. Differentiate between organized and unorganized drug.
- g.) What do you mean by adulteration? Explain types of adulteration with example.
- h.) Describe antiviral & anticancer drugs from marine source.
- i.) Write biological source, Chemical constituent and uses of any two of the following.
 - a) Gelatin
 - b) Serratiopeptidase
 - c) Agar

**Dr BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY,**

LONERE - RAIGAD - 402 103

, Winter Semester Examination: Dec. - 2019

Branch: B. Pharmacy Second Year

Semester-IV

Sub:PHARMACOGNOSY AND PHYTOCHEMISTRY - I [BP405T]

Date: 13/12/2019

Marks: 75

Time: 3 Hrs.

Instructions: i) All questions are compulsory
ii) Figures to the right indicate full marks.

ii) Figures to the right indicate full marks.

Q.I. Attempt following Multiple Choice Questions. [1×20=20]

1. coined the term Pharmacognosy.

- a) Theophrastus
b) Gallen
c) Pelletier
d) Seydler

2. Hippocrates is also known as “.....”

- a) Father of Medicine b) Father of Surgery
c) Father of Diagnosis d) — Father of

Pharmacognosy

3. Part of plant used for initiation of culture is called as.....

- a) Callus b) Explant
c) Scion d) Stock

4. One milligram of lycopodium spores contains.....spores.

- a) 92,000 b) 93,000
c) 94,000 d) 95,000

5. Anisocytic stomata are having number of Guard cells?

- a) one
b) two
c) three
d) four

6. The compound Cycloudesmol obtained from marine algae is mainly used as.....

- a) Anti-inflammatory
b) Cytotoxic
c) Cardio-active
d) Antibiotic

7. Sexual method of propagation includes.....

- a) Seed propagation
- b) Root propagation
- c) Rhizome propagation
- d) None of these

8.is responsible for ripening of fruits.

- a) IAA
- b) Ethylene
- c) 6-BA
- d) Absciscic acid

9. The temperature required for the cultivation of Cinchona is..... °F.

- a) 45 - 60
- b) 60 - 75
- c) 75 - 90
- d) 90 - 105

10. Cultivation ensures.....

- a) Irregular supply of drug
- b) Quality of drugs
- c) Quality and purity of drug
- d) All of above

11. Water insoluble portion of gum tragacanth is known as.....

- a) Tragacanthin
- b) Bassorin
- c) Inotragacanthin
- d) Guranin

12. Gum acacia is obtained from which plant?

- a) *Acacia arabica*
- b) *Acacia urens*
- c) *Acacia ribosa*
- d) *Acacia amansii*

13. is used as carbon source in Plant tissue culture medium.

- a) Sodium citrate
- b) Agar
- c) Disodium EDTA
- d) Sucrose

14. Alkaloids gives coloured precipitate along with Mayer's reagent.

- a) Reddish brown
- b) Reddish orange
- c) Yellow
- d) Cream

15. Molecular formula for isoprene subunit is.....

- a) C_4H_8
- b) C_5H_8
- c) C_6H_8
- d) C_7H_8

16. As per the Unani system of medicine **Al-Nar** represents the.....

- a) Water
- b) Fire
- c) Air
- d) Earth

17.is the optimum pH range of culture medium for growth of plant tissue.

- a) 4.5 to 5.0
- b) 5.0 to 5.5
- c) 5.5. to 6.0
- d) 6.0 to 6.5

18. Borntrager's test is positive only in case of type of anthraquinone glycosides.

- a) -C-type
- b) -O-type
- c) -S-type
- d) -N-type

19. Concept of **Triguna** is described in which system of medicine?

- a) Ayurveda
- b) Unani
- c) Siddha
- d) Chinese

20. Sting of bees and hornets are example of _____ class allergen.

- a) Ingestant
- b) Injectant
- c) Infectant
- d) Inhalant

Q.II. Answer any TWO of the following.

[10x2=20]

1) Differentiate between primary metabolites and secondary metabolites. What are alkaloids? Explain chemical classification of alkaloids with suitable examples and Write various preliminary phytochemical tests for detection of alkaloids.

2) Define plant tissue culture. Write detail note on composition of culture media. Enlist and explain various types of cultures. Write note on applications of plant tissue culture in Pharmacognosy.

3) What is the need of classification of crude drugs? Illustrate Alphabetical, Morphological, Chemical and Pharmacological classification of crude drugs.

Q.III. Answer any SEVEN of the following.

[05x07=35]

1) Describe in detail various sources of crude drugs and differentiate in between organized and unorganized crude drugs.

- 2) What do you mean by plant growth regulators? Explain various plant growth regulators along with suitable examples.
- 3) Discuss in detail factors affecting cultivation of medicinal and aromatic plants.
- 4) Illustrate theory and basic concepts, diagnosis and treatment aspects of Ayurvedic system of Medicine. Add note on role of pharmacognosy in alternative system of medicine.
- 5) Classify resins and tannins with suitable examples and give chemical tests for detection of glycosides.
- 6) What do you mean by edible vaccines? Enlist differences between conventional vaccines and edible vaccines? Add note on methods of transformation and mode of action of edible vaccines.
- 7) Describe in detail microscopic methods of evaluation of crude drugs.
- 8) Give biological source, method of preparation, chemical constituents, chemical tests and uses for following drugs
 - a) Agar
 - b) Acacia
- 9) Give the broad classification of marine drugs and explain each class in detail with suitable example.

●●●Paper End●●●



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – Summer 2022

Course: B. Pharmacy

Sem : IV

Subject Name: Pharmacognosy & Phytochemistry – I

Subject Code: BP405T

Max Marks: 75

Date: 08/09/2022

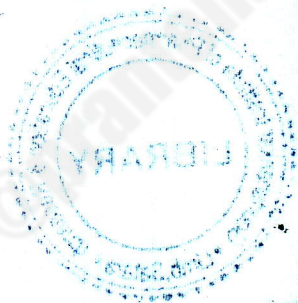
Duration: 3.45 Hr.

Instructions –

- All questions are compulsory
- Answers to MCQs should be written in full sentences
- Draw diagrams / figures wherever necessary
- Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = $20 \times 1 = 20$ (All the questions are compulsory)

1. Which of these is an organized drug?
a) Senna b) Gelatin c) Papain d) Benzoin
2. Glycosides are condensation products of
a) Sugar+protein b) Sugar+ lipid c) Sugar+agyclone d) Sugar+alkaloid
3. Stomatal number is number of stomata per----- of epidermis of leaf.
a) sq.m b) sq.mm c) sq.km d) sq.cm
4. Size of lycopodium spore is _____
a) 20 micron b) 30 micron c) 25 micron d) 35 micron
5. _____ is considered as father of medicine.
a) Hippocrates b) Aristotle c) Dioscorides d) Charaka
6. Taste and colour of agar comes under _____ evaluation of drug.
a) Chemical b) Biological c) Organoleptic d) Physical
7. Hemp contains



- a) High cellulose and low protein b) High cellulose and low lignin c) High lignin and low cellulose d) High lignin and low protein

8. Acid value is defined as

- a) number of kilogram of KOH required to neutralise 1 g of fat or oil
b) number of milligram of KOH required to neutralise 1 g of fat or oil
c) number of gram of KOH required to neutralise 1 g of fat or oil
d) number of ml of KOH required to neutralise 1 g of fat or oil

9. _____ belongs to the family of Apidae.

- a) Agar b) Acacia c) Tragacanth d) Honey

10. Waxes are example of.....

- a) Simple lipids b) Compound lipids c) Derived lipids d) Prostaglandins

11. Goldbeaters skin test is used for identification of.....

- a) Carbohydrates b) Lipids c) Tannins d) Glycosides

12. Tissue Culture medium should not be containing-----

- a) Pesticides b) Inorganic salts c) Vitamins d) Growth regulators

13. Indian origin therapy is

- a) Homeopathy b) Unani c) Allopathy d) Ayurveda

14. _____ oil is having laxative property

- a) Arachis oil b) Sesame oil c) Castor oil d) Corn oil

15. One of the four humours is

- a) White bile b) Phlegm c) Air d) Water

16. Fabric is prepared from.....

- a) Chitin b) Cellulose c) Hemicellulose d) None of these

17. Citral gum is used as adulterant of.....

- a) Acacia b) Agar c) Gelatin d) Tragacanth

18. Siddha medicine system originated from

- a) Karnataka b) Kerala c) Tamil Nadu d) Telangana

19. Sowing of rhizomes is

- a) Vegetative propagation b) seed propagation
c) sexual propagation d) Tissue culture

20. Bassorin is.....

- a) Water soluble b) Alcohol soluble c) Alcohol insoluble d) Water insoluble

Q. 2. Long Answers:

2 x 10 = 20 (Answer 2 out of 3)

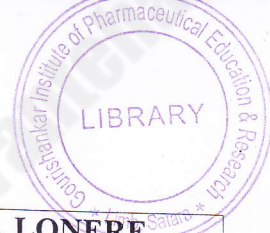
- a) What is cultivation? Explain the factors affecting cultivation of crude drugs.
b) Define Evaluation. Explain microscopic evaluation.
c) Define tissue culture. Explain the steps involved in Tissue culture. Mention its application.

Q. 3. Short Answers:

7 x 5 = 35 (Answer 7 out of 9)

- a) Define, classify and mention the identification tests for alkaloids.
b) Write the principle, diagnosis and treatment involved in Ayurveda.
c) What is Pharmacological classification of crude drugs? Mention its merits and demerits with examples.
d) State the biological source, chemical constituents of Tragacanth and Honey
e) Illustrate the pharmacognostic scheme on Cotton.
f) Differentiate between organized and unorganized crude drugs.
g) Explain the concept of polyploidy. Mention its applications.
h) Write a note on edible vaccines.
i) Define Pharmagony. Explain its scope and development.

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 16/02/2023

Course :	B. Pharmacy	Sem:	IV
Subject Name :	Pharmacognosy & Phytochemistry-I	Subject Code :	BP405T
Max Marks :	75	Duration :	3 Hr.

Instructions:

- All questions are compulsory
- Draw diagrams / figures wherever necessary
- Figures to right indicate full marks

Q. 1.	Objective Type Questions (Answer all the questions)	(10 x 2) = 20
i)	Define. a) Pharmacognosy b) Phytochemistry.	
ii)	Enlist sources of crude drugs with examples.	
iii)	Define a) Dried extracts b) Oleoresins.	
iv)	Write any two general identification tests of Alkaloids.	
v)	Write Biological source and uses of Honey.	
vi)	Define the term allergens with examples.	
vii)	What are Secondary Metabolites?	
viii)	Mention general properties of Volatile oils.	
ix)	Write Chemical constituents and uses of Wool fat.	
x)	Explain the term Plant Hormones.	
Q. 2.	Long Answers (Answer 2 out of 3)	(2 x 10) = 20
i)	Explain different methods of classification of crude drugs with examples.	
ii)	Define Evaluation of crude drugs. Explain Physical evaluation in detail.	
iii)	Write the principle, diagnosis and treatment involved in Ayurveda and Homeopathy system of medicine.	
Q. 3.	Short Answers (Answer 7 out of 9)	(7 x 5) = 35
i)	Define, classify and write general properties of Glycosides.	
ii)	Enlist factors affecting Cultivation. Write a note on edaphic factors	
iii)	Explain the concept of Hybridization. Mention its applications.	
iv)	Explain the steps involved in Plant Tissue culture technique.	
v)	What is quantitative microscopy? Write a note on Lycopodium spore method.	
vi)	Differentiate between organized and unorganized crude drugs.	
vii)	State biological source and chemical constituents of Castor oil and Acacia.	
viii)	Write biological source and preparation method of Gelatin and Papain.	
ix)	Give pharmacognostic account of Cotton.	

Course: B. Pharmacy

Subject Name: Pharmacognosy & Phytochemistry I

Max Marks: 75

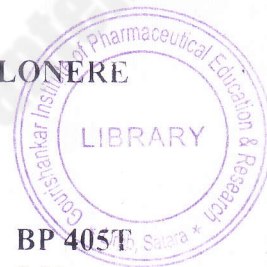
Sem:IV

Subject Code:

Duration:

BP 405T

3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Define Pharmacognosy and give an example of organized and unorganized crude drug.
- ii) Define mutation and enlist types of mutation.
- iii) Give the biological source and uses of Castor oil.
- iv) Define Teratogen and give an example of Teratogen.
- v) Enlist Crude drug evaluation methods (Any four).
- vi) What are glycosides? Give an example of a drug containing glycosides.
- vii) What is the function of Auxin in the development of plants?
- viii) Give any two examples of crude drugs derived from animal origin.
- ix) What are primary metabolites? Give any two examples of it.
- x) Write contribution of following scientists to Pharmacognosy.
a) Charak b) Seydler

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Define Adulteration and explain various types of adulteration of crude drugs with suitable examples.
- ii) Illustrate theory, basic concept, diagnosis, treatment and dosage forms used in Ayurvedic system of medicine.
- iii) What are plant hormones? Explain various plant hormones along with their functions and examples.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Discuss in detail about extrinsic and intrinsic factors affecting the cultivation of medicinal and aromatic plants. (give details of any 4 factors)
- ii) Give the biological source, method of preparation, and uses of following drugs.

a) Gelatin b) Papain

- iii) Define plant tissue culture and explain any four types of plant tissue cultures.
- iv) Write a note on Edible vaccines.
- v) Explain historical development of Pharmacognosy.
- vi) Define and classify Alkaloids. Give the properties, identification test for the same.
- vii) Give the pharmacognostic account of Chaulmoogra oil.
- viii) Define Hallucinogens and natural allergens. Classify the natural allergens along with suitable examples.
- ix) What is the need for classification of crude drugs? Explain morphological classification of crude drugs, its advantages and limitations.

-----END OF THE PAPER-----

Date: 11/01/2024

Course : B. Pharmacy

Sem: IV

Subject Name : Pharmacognosy & Phytochemistry I

Subject Code :BP405T

Max Marks : 75

Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) What are alkaloids
- ii) Enlist the different methods and significance of drying method?
- iii) What are edible vaccines?
- iv) Define pharmacognosy. Who coined the term pharmacognosy?
- v) Mention any two drugs from Protein and proteolytic enzyme category.
- vi) Define Stomatal Number with Ex.
- vii) Explain any two chemical tests for Flavonoids.
- viii) Explain the MOA of Gibberellins
- ix) State any four Traditional Chinese Medicine therapies.
- x) State the biological source of cotton?

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Classify the crude drugs on the basis of their morphology, Pharmacological activity with numerous examples.
- ii) What are the methods of adulterations involved in crude drug? Write a detailed note on physical evaluation of crude drugs.
- iii) What is PTC? Discuss the various nutritional requirements of plant tissue culture?

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Explain the Ayurvedic system of medicine and its role in pharmacognosy.
- ii) Discuss pharmacognostic profile of Tragacanth
- iii) Define Tannins? Give its classification
- iv) Explain novel medicinal plants from marine sources
- v) Explain the specific chemical test for Anthraquinone and Cardiac glycosides.
- vi) Write the short note on Natural allergens.
- vii) Define Alkaloids and Give its classification and identification tests.
- viii) Define Ash value. Explain the determination of Ash value
- ix) Write the biological source, chemical constituents and uses of bees wax and honey

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Subject Name: Pharmacognosy & Phytochemistry-I

Max Marks: 75

Date: 25-06-2024

Semester: IV Semester

Subject Code: BP405T

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10 X 2) = 20

- i) Define a) Pharmacognosy b) Phytochemistry.
- ii) What are Primary Metabolites?
- iii) Write any four general properties of Volatile oil.
- iv) Explain two chemical tests for Tannins.
- v) State biological source and uses of Acacia.
- vi) What are edible vaccines?
- vii) Differentiate between organised & unorganized crude drugs.
- viii) Define a) Teratogens b) Hallucinogens.
- ix) Write the uses of Pepsin & Wool Fat.
- x) What is Polyploidy?

Q.2. Long Answers (Answer 2 out of 3) (10 X 2) = 20

- i) Enlist methods of classification of crude drugs. Write a detailed note on Morphological and Chemical classification of crude drugs with examples.
- ii) Define Plant Tissue Culture Technique. Explain different steps of PTC with its applications in Pharmacognosy.
- iii) Discuss the different methods used in adulterations of crude drugs. Write a detailed note on Microscopical evaluation of crude drugs.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Explain different factors affecting the cultivation.
- ii) Write a note on Plant Growth Regulators.
- iii) Define Glycosides? Give its classification.
- iv) Write the biological source, chemical constituents and uses of Cotton and Agar.
- v) Give an account on Marine Drugs.
- vi) Write history and scope of Pharmacognosy.
- vii) Write Principle, diagnosis and treatment of Homeopathic system of Medicine.
- viii) Write a note on Natural allergens.
- ix) Give pharmacognostic account of Castor oil.

***** END OF THE PAPER *****

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